Time Series Data Analysis and Forecasting

MSDS

Module 4

Topic Covered

- Exponential Smoothing
- Exponential smoothing methods
- Autoregressive (AR) models
- Moving average (MA) models
- Integrated (I) models
- Introduction to ARIMA models
- Ljung-Box test
- Use cases

Exponential Smoothing

- Exponential smoothing was proposed in late 1950s (Brown, 1959; Holt, 1957; Winters, 1960)
- Forecasts produced using exponential smoothing methods are **weighted averages of past observations**, with weights decaying exponentially as observations get older.
- the more recent the observation, higher the associated weight.
- It generates reliable forecasts quickly and for a wide range of time series
- Provides major importance to applications in industry.
- selection of method is based on recognising key components of time series (**trend and seasonal)**
- Also, the way in which these enter smoothing method (e.g., in **an additive, damped or multiplicative manner**).

Simple Exponential Smoothing

- This method is suitable for forecasting data with **no clear trend or seasonal pattern**
- As the data in Figure**, do not display any clear trending** behavior or any seasonality.
- It shows is a rise in the last few years, which might suggest a trend.
- Using **naïve method**, all forecasts for future are **equal to last observed value** of series Or average method is used to forecast values

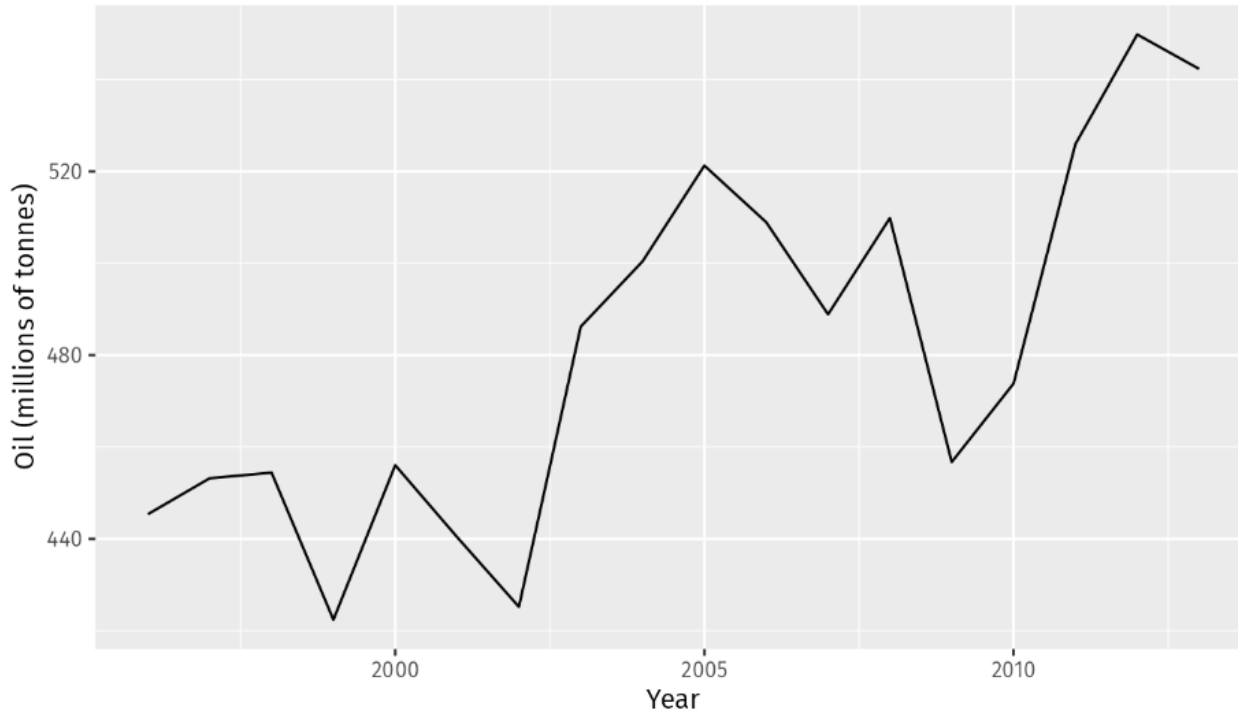


Figure 7.1: Oil production in Saudi Arabia from 1996 to 2013.

Naïve Method

$$\hat{y}_{T+h|T} = y_T$$

$$\text{for } h = 1, 2, \ldots.$$

Average Method

$$\hat{y}_{T+h|T} = \frac{1}{T}\sum_{t=1}^{T} y_t,$$

Simple Exponential Smoothing

- As both naïve method or average method does not associate any weight to past values Weighted average method is used to make forecast better and more accurate.
- Attach **larger weights to more recent observations** than to observations from distant past
- In exponential method, Forecasts are calculated using **weighted averages**, where the weights decrease exponentially as observations come from further in the past — the smallest weights are associated with the oldest observations.

$$\hat{y}_{T+1|T} = \alpha y_T + \alpha(1-\alpha)y_{T-1} + \alpha(1-\alpha)^2 y_{T-2} + \cdots,$$

where $0 \leq \alpha \leq 1$ is the smoothing parameter. The one-step-ahead forecast for time $T+1$ is a weighted average of all of the observations in the series $y_1, \ldots, y_T$. The rate at which the weights decrease is controlled by the parameter α.

Simple Exponential Smoothing

- table shows the weights attached to observations for four different values of α when forecasting using simple exponential smoothing.
- Note that sum of the weights even for a small value of α will be approximately one for any reasonable sample size.
- For any **α between 0 and 1**, weights attached to observations decrease exponentially as we go back in time, hence name **"exponential smoothing"**

	$\alpha = 0.2$	$\alpha = 0.4$	$\alpha = 0.6$	$\alpha = 0.8$
y_T	0.2000	0.4000	0.6000	0.8000
y_{T-1}	0.1600	0.2400	0.2400	0.1600
y_{T-2}	0.1280	0.1440	0.0960	0.0320
y_{T-3}	0.1024	0.0864	0.0384	0.0064
y_{T-4}	0.0819	0.0518	0.0154	0.0013
y_{T-5}	0.0655	0.0311	0.0061	0.0003

Two Forms of Simple Exponential Smoothing

- Weighted average form
- Component form

Weighted average form for Simple Exponential Smoothing

The forecast at time $T+1$ is equal to a weighted average between the most recent observatio
y_T and the previous forecast $\hat{y}_{T|T-1}$:

$$\hat{y}_{T+1|T} = \alpha y_T + (1-\alpha)\hat{y}_{T|T-1},$$

where $0 \leq \alpha \leq 1$ is the smoothing parameter. Similarly, we can write the fitted values as

$$\hat{y}_{t+1|t} = \alpha y_t + (1-\alpha)\hat{y}_{t|t-1},$$

for $t = 1, \ldots, T$. (Recall that fitted values are simply one-step forecasts of the training data.)

$$\begin{aligned}
\hat{y}_{2|1} &= \alpha y_1 + (1-\alpha)\ell_0 \\
\hat{y}_{3|2} &= \alpha y_2 + (1-\alpha)\hat{y}_{2|1} \\
\hat{y}_{4|3} &= \alpha y_3 + (1-\alpha)\hat{y}_{3|2} \\
&\vdots \\
\hat{y}_{T|T-1} &= \alpha y_{T-1} + (1-\alpha)\hat{y}_{T-1|T-2} \\
\hat{y}_{T+1|T} &= \alpha y_T + (1-\alpha)\hat{y}_{T|T-1}.
\end{aligned}$$

$$\begin{aligned}
\hat{y}_{3|2} &= \alpha y_2 + (1-\alpha)\left[\alpha y_1 + (1-\alpha)\ell_0\right] \\
&= \alpha y_2 + \alpha(1-\alpha)y_1 + (1-\alpha)^2\ell_0 \\
\hat{y}_{4|3} &= \alpha y_3 + (1-\alpha)[\alpha y_2 + \alpha(1-\alpha)y_1 + (1-\alpha)^2\ell_0] \\
&= \alpha y_3 + \alpha(1-\alpha)y_2 + \alpha(1-\alpha)^2 y_1 + (1-\alpha)^3\ell_0 \\
&\vdots \\
\hat{y}_{T+1|T} &= \sum_{j=0}^{T-1} \alpha(1-\alpha)^j y_{T-j} + (1-\alpha)^T \ell_0.
\end{aligned}$$

Component form for Simple Exponential Smoothing

- This representations of exponential smoothing methods comprise a forecast equation and a smoothing equation for each of the components included in the method.
- The form for simple exponential smoothing is given by:

$$\begin{aligned} \text{Forecast equation} \qquad & \hat{y}_{t+h|t} = \ell_t \\ \text{Smoothing equation} \qquad & \ell_t = \alpha y_t + (1-\alpha)\ell_{t-1}, \end{aligned}$$

where ℓ_t is the level (or the smoothed value) of the series at time t. Setting $h = 1$ gives the fitted values, while setting $t = T$ gives the true forecasts beyond the training data.

Example: Oil Production

Simple exponential smoothing is applied to forecast oil production in Saudi Arabia.

```
oildata <- window(oil, start=1996)
# Estimate parameters
fc <- ses(oildata, h=5)
# Accuracy of one-step-ahead training errors
round(accuracy(fc),2)
#>               ME  RMSE   MAE MPE MAPE MASE  ACF1
#> Training set 6.4 28.12 22.26 1.1 4.61 0.93 -0.03
```

This gives parameter estimates $\hat{\alpha} = 0.83$ and $\hat{\ell}_0 = 446.6$, obtained by minimising SSE over periods $t = 1, 2, \ldots, 18$, subject to the restriction that $0 \leq \alpha \leq 1$.

In Table 7.1 we demonstrate the calculation using these parameters. The second last column shows the estimated level for times $t = 0$ to $t = 18$; the last few rows of the last column show the forecasts for $h = 1, 2, 3, 4, 5$.

Example: Oil Production

- **second last column** shows the estimated level for times t=0 to t=18
- the **last few rows** of the last column show the forecasts for h=1,2,3,4,5.
- Forecasting the **total oil production** in millions of tonnes for Saudi Arabia

Year	Time	Observation	Level	Forecast
	t	y_t	ℓ_t	$\hat{y}_{t\|t-1}$
1995	0		446.59	
1996	1	445.36	445.57	446.59
1997	2	453.20	451.93	445.57
1998	3	454.41	454.00	451.93
1999	4	422.38	427.63	454.00
2000	5	456.04	451.32	427.63
2001	6	440.39	442.20	451.32
2002	7	425.19	428.02	442.20
2003	8	486.21	476.54	428.02
2004	9	500.43	496.46	476.54
2005	10	521.28	517.15	496.46
2006	11	508.95	510.31	517.15
2007	12	488.89	492.45	510.31
2008	13	509.87	506.98	492.45
2009	14	456.72	465.07	506.98
2010	15	473.82	472.36	465.07
2011	16	525.95	517.05	472.36
2012	17	549.83	544.39	517.05
2013	18	542.34	542.68	544.39
	h			$\hat{y}_{T+h\|T}$
2014	1			542.68
2015	2			542.68
2016	3			542.68
2017	4			542.68
2018	5			542.68

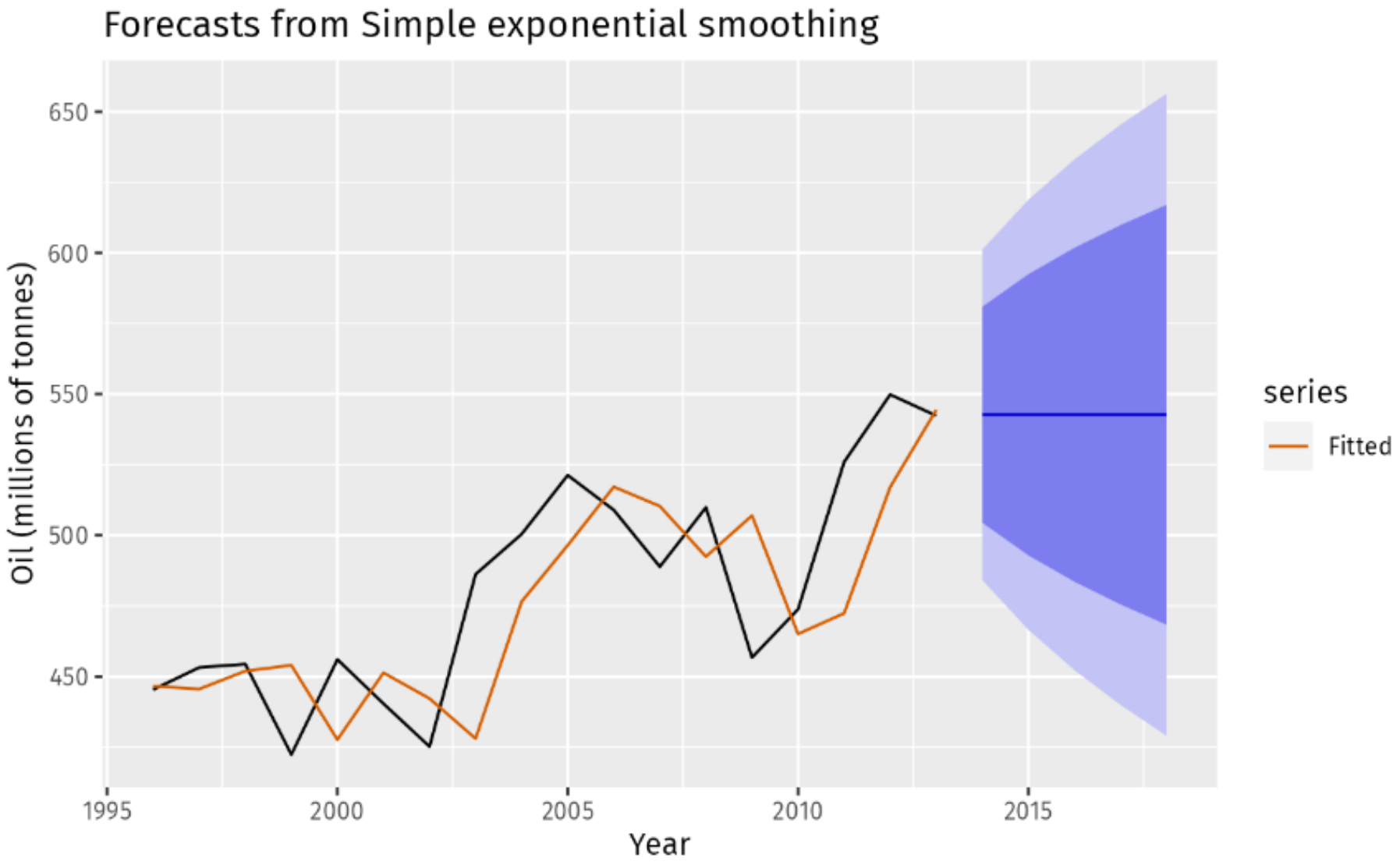


Figure 7.2: Simple exponential smoothing applied to oil production in Saudi Arabia (1996–2013).

Holt's Linear trend method

- Holt (1957) extended simple exponential smoothing to allow forecasting of data **with a trend**.
- This method involves a **forecast equation** and **two smoothing equations** (one for the level and one for the trend

$$
\begin{aligned}
\text{Forecast equation} \quad & \hat{y}_{t+h|t} = \ell_t + hb_t \\
\text{Level equation} \quad & \ell_t = \alpha y_t + (1-\alpha)(\ell_{t-1} + b_{t-1}) \\
\text{Trend equation} \quad & b_t = \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)b_{t-1}
\end{aligned}
$$

where ℓ_t denotes an estimate of the level of the series at time t, b_t denotes an estimate of the trend (slope) of the series at time t, α is the smoothing parameter for the level, $0 \leq \alpha \leq 1$, and β^* is the smoothing parameter for the trend, $0 \leq \beta^* \leq 1$. (We denote this as β^* instead of β for

Damped trend methods

- forecasts generated by Holt's linear method display a **constant trend** (increasing or decreasing) indefinitely into future.
- Empirical evidence indicates that these methods tend to **over-forecast**, especially for longer forecast horizons. Motivated by this observation, Gardner & McKenzie (1985) introduced a parameter that **"dampens"** the trend to a flat line some time in the future.
- Methods that include a damped trend have proven to be very successful, and are arguably most popular individual methods when forecasts are required **automatically for many series**.

Damped trend methods

In conjunction with the smoothing parameters α and β^* (with values between 0 and 1 as in Holt's method), this method also includes a damping parameter $0 < \phi < 1$:

$$\begin{aligned}
\hat{y}_{t+h|t} &= \ell_t + (\phi + \phi^2 + \cdots + \phi^h) b_t \\
\ell_t &= \alpha y_t + (1-\alpha)(\ell_{t-1} + \phi b_{t-1}) \\
b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)\phi b_{t-1}.
\end{aligned}$$

If $\phi = 1$, the method is identical to Holt's linear method. For values between 0 and 1, ϕ dampens the trend so that it approaches a constant some time in the future. In fact, the forecasts converge to $\ell_T + \phi b_T/(1-\phi)$ as $h \to \infty$ for any value $0 < \phi < 1$. This means that short-run forecasts are trended while long-run forecasts are constant.

Holt's Winters Seasonal Method

- Holt (1957) and Winters (1960) extended Holt's method to capture seasonality.
- It comprises the forecast equation and three smoothing equations — one for the level **ℓt**, one for the trend **bt**, and one for the seasonal component **st**, with corresponding smoothing parameters **α, β*** and **γ**.
- We use ***m*** to denote frequency of seasonality (number of seasons in a year)
- For example, for quarterly data m=4, and for monthly data m=12.

Holt-Winters' additive method

The component form for the additive method is:

$$
\begin{aligned}
\hat{y}_{t+h|t} &= \ell_t + hb_t + s_{t+h-m(k+1)} \\
\ell_t &= \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}) \\
b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)b_{t-1} \\
s_t &= \gamma(y_t - \ell_{t-1} - b_{t-1}) + (1-\gamma)s_{t-m},
\end{aligned}
$$

where k is the integer part of $(h-1)/m$, which ensures that the estimates of the seasonal indices used for forecasting come from the final year of the sample. The level equation shows a weighted average between the seasonally adjusted observation $(y_t - s_{t-m})$ and the non-seasonal forecast $(\ell_{t-1} + b_{t-1})$ for time t. The trend equation is identical to Holt's linear method. The seasonal equation shows a weighted average between the current seasonal index, $(y_t - \ell_{t-1} - b_{t-1})$, and the seasonal index of the same season last year (i.e., m time periods ago).

Holt-Winters' multiplicative method

The component form for the multiplicative method is:

$$
\begin{aligned}
\hat{y}_{t+h|t} &= (\ell_t + hb_t)s_{t+h-m(k+1)} \\
\ell_t &= \alpha \frac{y_t}{s_{t-m}} + (1-\alpha)(\ell_{t-1} + b_{t-1}) \\
b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1-\beta^*)b_{t-1} \\
s_t &= \gamma \frac{y_t}{(\ell_{t-1} + b_{t-1})} + (1-\gamma)s_{t-m}
\end{aligned}
$$

Example- Holts winters additive seasonality

Forecast for austourists dataset

t	y_t	ℓ_t	b_t	s_t	$\hat{y}_t$	
2004 Q1	-3				9.70	
2004 Q2	-2				-9.31	
2004 Q3	-1				-1.69	
2004 Q4	0		32.26	0.70	1.31	
2005 Q1	1	42.21	32.82	0.70	9.50	42.66
2005 Q2	2	24.65	33.66	0.70	-9.13	24.21
2005 Q3	3	32.67	34.36	0.70	-1.69	32.67
2005 Q4	4	37.26	35.33	0.70	1.69	36.37
	⋮	⋮	⋮	⋮	⋮	⋮
2015 Q1	41	73.26	59.96	0.70	12.18	69.05
2015 Q2	42	47.70	60.69	0.70	-13.02	47.59
2015 Q3	43	61.10	61.96	0.70	-1.35	59.24
2015 Q4	44	66.06	63.22	0.70	2.35	64.22
	h					$\hat{y}_{T+h\|T}$
2016 Q1	1					76.10
2016 Q2	2					51.60
2016 Q3	3					63.97
2016 Q4	4					68.37
2017 Q1	5					78.90
2017 Q2	6					54.41
2017 Q3	7					66.77
2017 Q4	8					71.18

Autoregressive model

- In a multiple regression model, we forecast variable of interest using a linear combination of predictors.
- In an autoregression model, we forecast variable of interest using a **linear combination of *past values*** *of the variable*.
- The term ***autoregression*** indicates that it is a regression of variable against itself.
- an autoregressive model of order **p** can be written as

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + \varepsilon_t,$$

where ε_t is white noise. This is like a multiple regression but with *lagged values* of y_t as predictors. We refer to this as an **AR(p) model**, an autoregressive model of order p.

Autoregressive model

- Autoregressive models are flexible at handling a wide range of time series patterns
- Figure show series from an AR(1) model and an AR(2) model

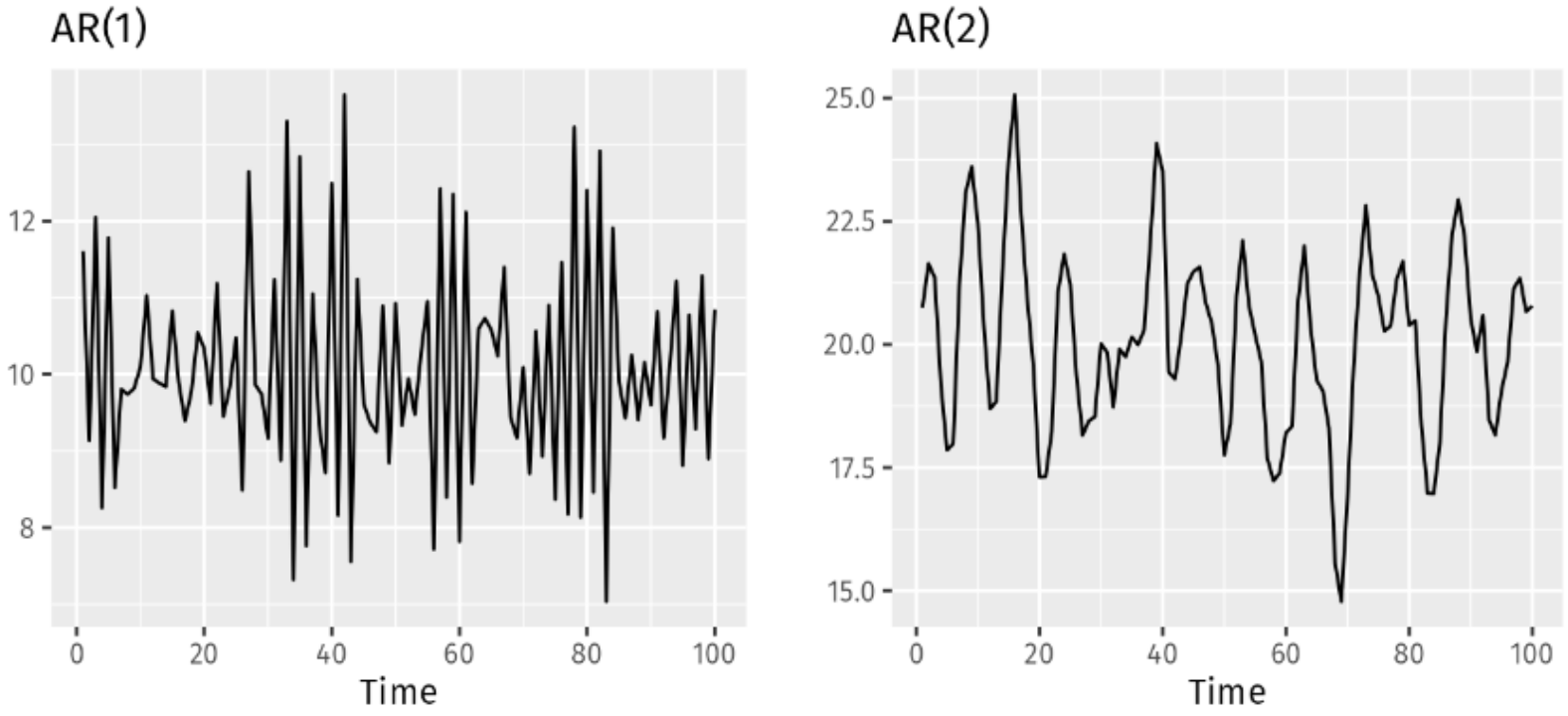


Figure 8.5: Two examples of data from autoregressive models with different parameters. Left: AR(1) with $y_t = 18 - 0.8y_{t-1} + \varepsilon_t$. Right: AR(2) with $y_t = 8 + 1.3y_{t-1} - 0.7y_{t-2} + \varepsilon_t$. In both cases, ε_t is normally distributed white noise with mean zero and variance one.

Autoregressive model

For an AR(1) model:

- when $\phi_1 = 0$, y_t is equivalent to white noise;
- when $\phi_1 = 1$ and $c = 0$, y_t is equivalent to a random walk;
- when $\phi_1 = 1$ and $c \neq 0$, y_t is equivalent to a random walk with drift;
- when $\phi_1 < 0$, y_t tends to oscillate around the mean.

We normally restrict autoregressive models to stationary data, in which case some constraints on the values of the parameters are required.

- For an AR(1) model: $-1 < \phi_1 < 1$.
- For an AR(2) model: $-1 < \phi_2 < 1$, $\phi_1 + \phi_2 < 1$, $\phi_2 - \phi_1 < 1$.

When $p \geq 3$, the restrictions are much more complicated. R takes care of these restrictions when estimating a model.

Moving average model(MA)

- Rather than using past values of the forecast variable in a regression, a moving average model uses **past forecast errors** in a regression-like model.

$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q},$$

where ε_t is white noise. We refer to this as an **MA(q) model**, a moving average model of order q. Of course, we do not *observe* the values of ε_t, so it is not really a regression in the usual sense.

- each value of y_t is thought of as a weighted moving average of past few forecast errors.
- However, **moving average *models*** should not be confused with **moving average *smoothing***
- A moving average model is used for **forecasting future values**, while moving average smoothing is used for **estimating the trend-cycle** of past values.

Moving average model(MA)

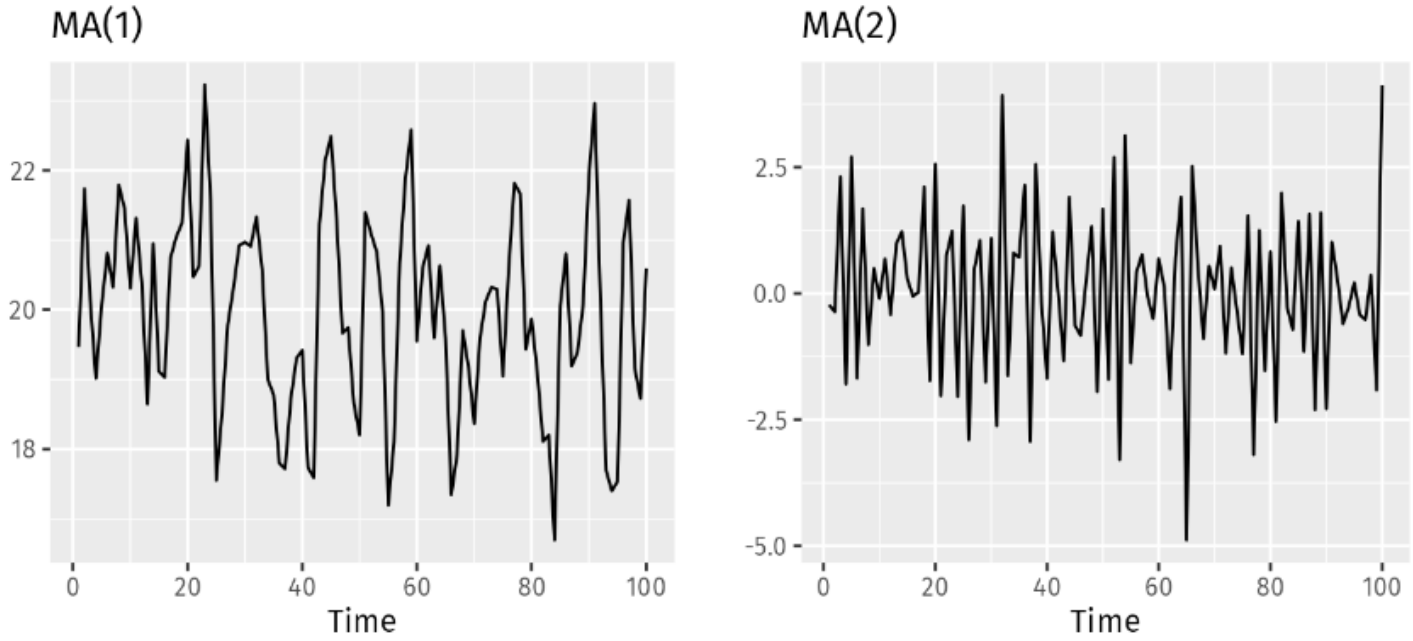


Figure 8.6: Two examples of data from moving average models with different parameters. Left: MA(1) with $y_t = 20 + \varepsilon_t + 0.8\varepsilon_{t-1}$. Right: MA(2) with $y_t = \varepsilon_t - \varepsilon_{t-1} + 0.8\varepsilon_{t-2}$. In both cases, ε_t is normally distributed white noise with mean zero and variance one.

ARIMA Model

- **ARIMA**, short for **AutoRegressive Integrated Moving Average**, is a forecasting algorithm based on the idea that the information in the past values of the time series can alone be used to predict the future values.
- Specified by three order parameters: (p, d, q),where,
  - p is the order of the AR term
  - q is the order of the MA term
  - d is the number of differencing required to make the time series stationary

Types of ARIMA Model

- **ARIMA** : Non-seasonal Autoregressive Integrated Moving Averages
- **SARIMA** : Seasonal ARIMA
- **SARIMAX** : Seasonal ARIMA with exogenous variables

If a time series, has seasonal patterns, then we need to add seasonal terms and it becomes SARIMA, short for **Seasonal ARIMA**.

Non Seasonal ARIMA Model

- If we combine differencing with **autoregression** and a **moving average model**, we obtain a non-seasonal ARIMA model.
- ARIMA is an acronym for Auto Regressive Integrated Moving Average ("integration" is reverse of differencing).
- The full model can be written as

$$y'_t = c + \phi_1 y'_{t-1} + \cdots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} + \varepsilon_t,$$

- The "predictors" on the right hand side include both lagged values of yt and lagged errors called as **ARIMA(p, d, q)** Model where,

p=the order of the regressive part
d=degree of the first differencing involved
q=order of the moving average part

Special cases ARIMA Model

White noise	ARIMA(0,0,0)
Random walk	ARIMA(0,1,0) with no constant
Random walk with drift	ARIMA(0,1,0) with a constant
Autoregression	ARIMA(p,0,0)
Moving average	ARIMA(0,0,q)

Once we start combining components in this way to form more complicated models, it is much easier to work with the backshift notation.
Equation in backshift notation is now:-

$$\underbrace{(1-\phi_1 B-\cdots-\phi_p B^p)}_{\mathrm{AR}(p)}\quad \underbrace{(1-B)^d}_{d\text{ differences}} y_t \quad = \quad c+(1+\underbrace{\theta_1 B+\cdots+\theta_q B^q}_{\mathrm{MA}(q)})\varepsilon_t$$

ARIMA Model in R

R uses a slightly different parameterisation:

$$(1 - \phi_1 B - \cdots - \phi_p B^p)(y'_t - \mu) = (1 + \theta_1 B + \cdots + \theta_q B^q)\varepsilon_t,$$

where $y'_t = (1 - B)^d y_t$ and μ is the mean of y'_t. To convert to the form given by (8.2), set $c = \mu(1 - \phi_1 - \cdots - \phi_p)$.

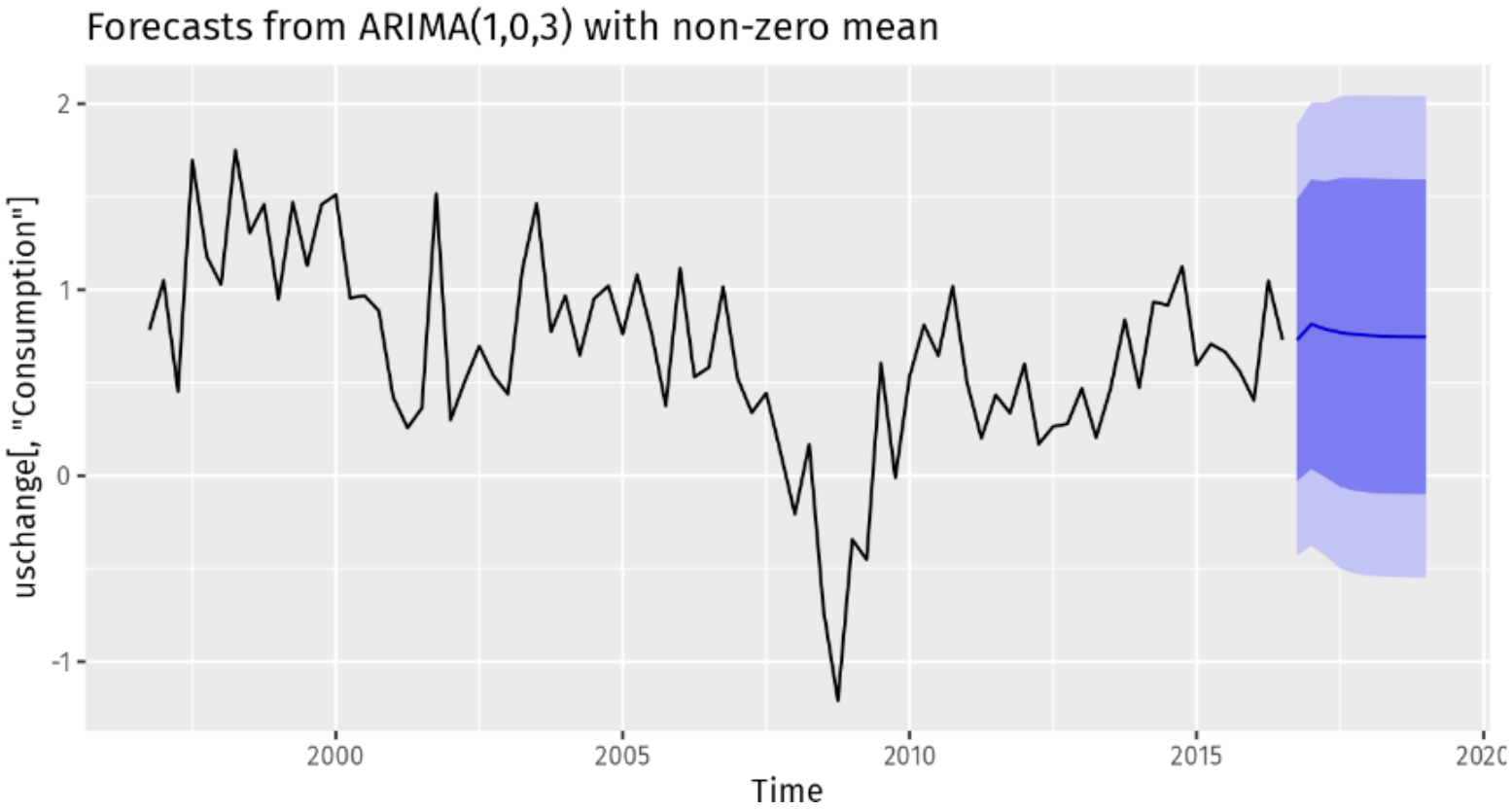
- Selecting values for p,d and q can be difficult.
- But, **auto.arima()** function in R does it for you automatically.

Understanding ARIMA Model(1)

- **auto.arima()** function is useful, but anything automated can be a little dangerous, and it is worth understanding something of the behaviour of the models
- The constant c has an important effect on the long-term forecasts obtained from these models.
- If **c=0 and d=0**, the long-term forecasts will go to zero.
- If **c=0 and d=1**, the long-term forecasts will go to a non-zero constant.
- If **c=0 and d=2**, the long-term forecasts will follow a straight line.
- If **c≠0 and d=0**, the long-term forecasts will go to the mean of the data.
- If **c≠0 and d=1**, the long-term forecasts will follow a straight line.
- If **c≠0 and d=2**, the long-term forecasts will follow a quadratic trend.

Understanding ARIMA Model(2)

- Value of d has an effect on prediction interval
- Higher the value of d , more rapidly prediction interval increase in size
- For d=0, long-term forecast standard deviation will go to the standard deviation of the historical data, so the prediction intervals will all be essentially the same.

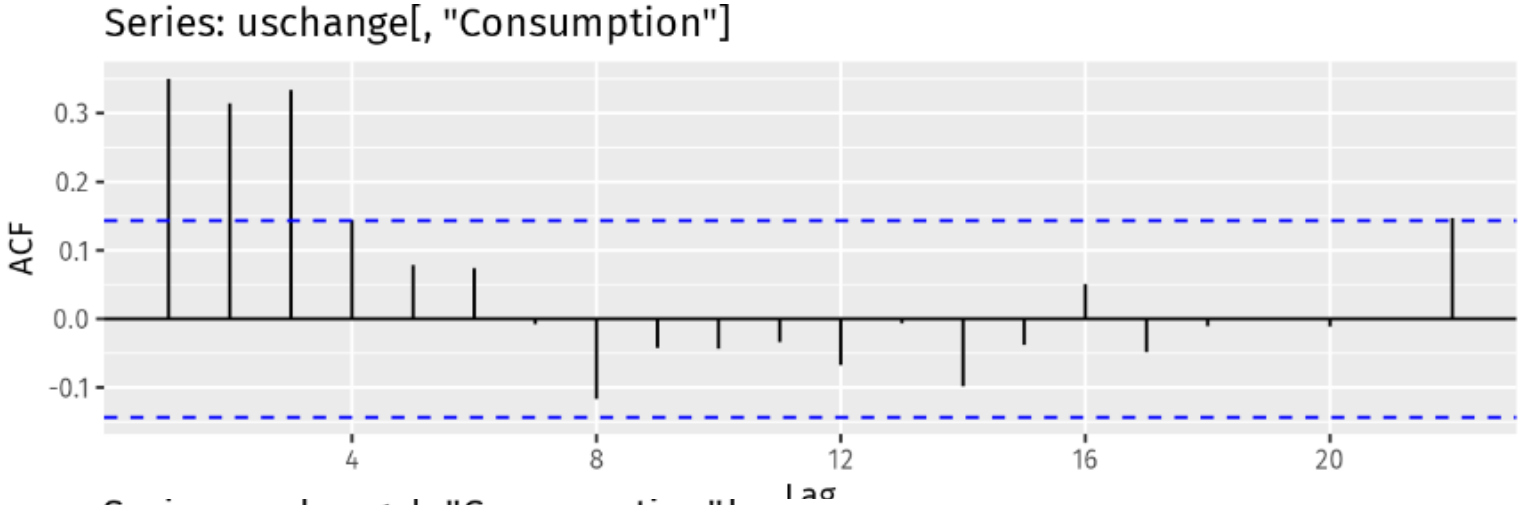


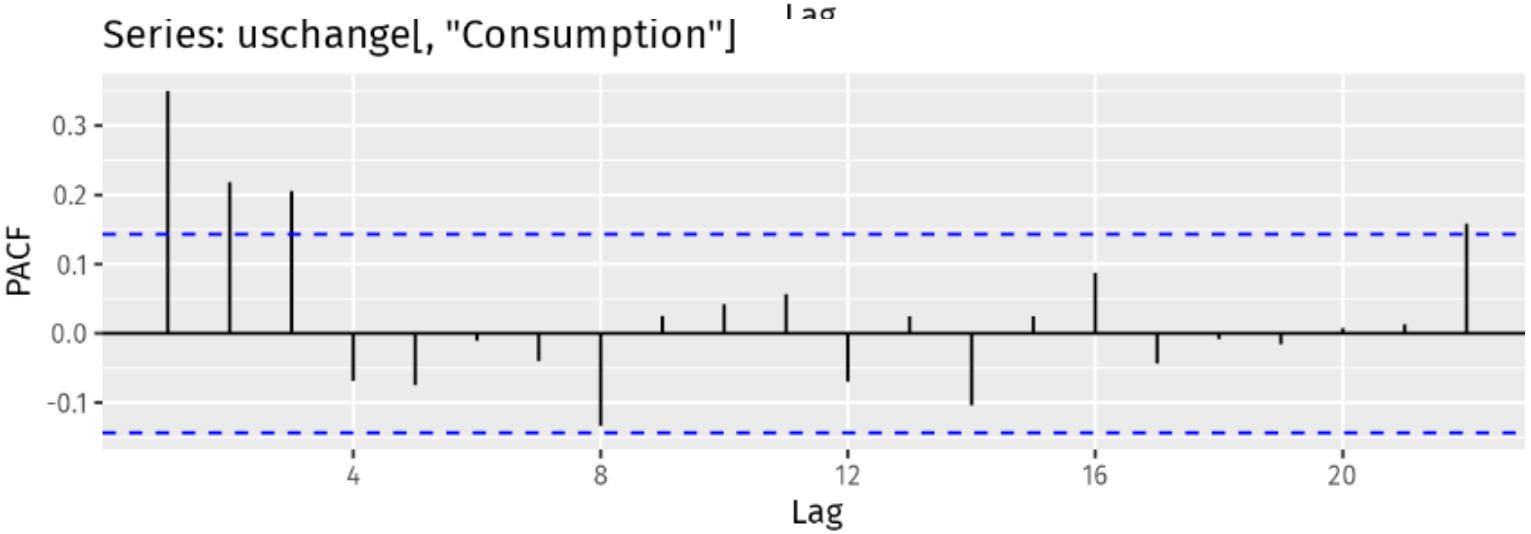
Understanding ARIMA Model(3)

- The value of **p** is important if the data show cycles.
- To obtain cyclic forecasts, it is necessary to have **p≥2**, along with some additional conditions on parameters
- It is usually not possible to tell, simply from a time plot, **what values of p and q** are appropriate for data.
- It is possible to use the ACF plot, and the closely related PACF plot, to determine appropriate values for p and q.

Showing ACF and PACF plot for US change data

Figures 8.9 and 8.10 shows the ACF and PACF plots for the US consumption data shown in Figure 8.7. The partial autocorrelations have the same critical values of $\pm 1.96/\sqrt{T}$ as for ordinary autocorrelations, and these are typically shown on the plot as in Figure 8.9.





- There are three spikes in the ACF, followed by an almost significant spike at lag 4.
- In PACF, there are three significant spikes, and then no significant spikes thereafter (apart from one just outside the bounds at lag 22).
- As per ACF and PACF, an ARIMA(3,0,0) model might be appropriate.

Choosing between ARIMA(p, d,0) and ARIMA(0,d,q)

- If the data are from an ARIMA(p,d,0) or ARIMA(0,d,q) model, then ACF and PACF plots can be helful in detrmining p and q.
- If p and q are both positive then plots do not help in finding suitable values of p and q.

Data may follow **ARIMA(p, d, 0)** if ACF and PACF plots of diffrenced data show patterns like:-

- ACF is exponentially decaying or sinusoidal
- There is significant spike at lag p in PACF but none beyond lag p.

Data may follow **ARIMA(0, d, q)** if ACF and PACF plots of diffrenced data show patterns like:-

- PACF is exponentially decaying or sinusoidal
- There is significant spike at lag q in ACF but none beyond lag q

Estimation and order selection

Maximum likelihood estimation-

- Once the model order has been identified (values of p, d and q), we need to estimate the parameters c, ϕ1,…,ϕp,…, θ1,…,θq.
- When R estimates the ARIMA model, it uses *maximum likelihood estimation* (MLE).
- This technique finds values of the parameters which maximise the probability of obtaining the data that we have observed.
- For ARIMA models, MLE is similar to the *least squares* estimates that would be obtained by minimizing,

$$\sum_{t=1}^{T} \varepsilon_t^2 .$$

Information Criteria

- Akaike's Information Criterion (AIC), which was useful in selecting predictors for regression, is also useful for determining the order of an ARIMA model.
- It can be written as

$$\text{AIC} = -2\log(L) + 2(p + q + k + 1),$$

- where L is the likelihood of the data, k=1 if c≠0 and k=0 if c=0.
- Note that the last term in parentheses is the number of parameters in the model (including σ2, variance of the residuals).

- For ARIMA models, the corrected AIC can be written as

$$\text{AICc} = \text{AIC} + \frac{2(p + q + k + 1)(p + q + k + 2)}{T - p - q - k - 2},$$

Bayesian Information Criterion

- **Bayesian information criterion** (**BIC**) or **Schwarz information criterion**
- It is a criterion for model selection among a finite set of models
- models with lower BIC are generally preferred

$$\mathrm{BIC} = \mathrm{AIC} + [\log(T) - 2](p + q + k + 1).$$

How does auto.arima() work?(1)

- auto.arima() function in R used a variation of the Hyndman-Khandakar algorithm
- Algorithm combines unit root tests, minimization of the AICc anf MLE to obtain an ARIMA model
- Argument to auto.arima() provide many variations on algorithms

Hyndman–Khandakar algorithm for automatic ARIMA modelling

1. The number of differences $0 \leq d \leq 2$ is determined using repeated KPSS tests.

2. The values of p and q are then chosen by minimising the AICc after differencing the data d times. Rather than considering every possible combination of p and q, the algorithm uses a stepwise search to traverse the model space.

   a. Four initial models are fitted:
      - $\text{ARIMA}(0, d, 0)$,
      - $\text{ARIMA}(2, d, 2)$,
      - $\text{ARIMA}(1, d, 0)$,
      - $\text{ARIMA}(0, d, 1)$.

      A constant is included unless $d = 2$. If $d \leq 1$, an additional model is also fitted:
      - $\text{ARIMA}(0, d, 0)$ without a constant.

How does auto.arima() work?(2)

b. The best model (with the smallest AICc value) fitted in step (a) is set to be the "current model".

c. Variations on the current model are considered:

- vary p and/or q from the current model by ± 1;
- include/exclude c from the current model.

The best model considered so far (either the current model or one of these variations) becomes the new current model.

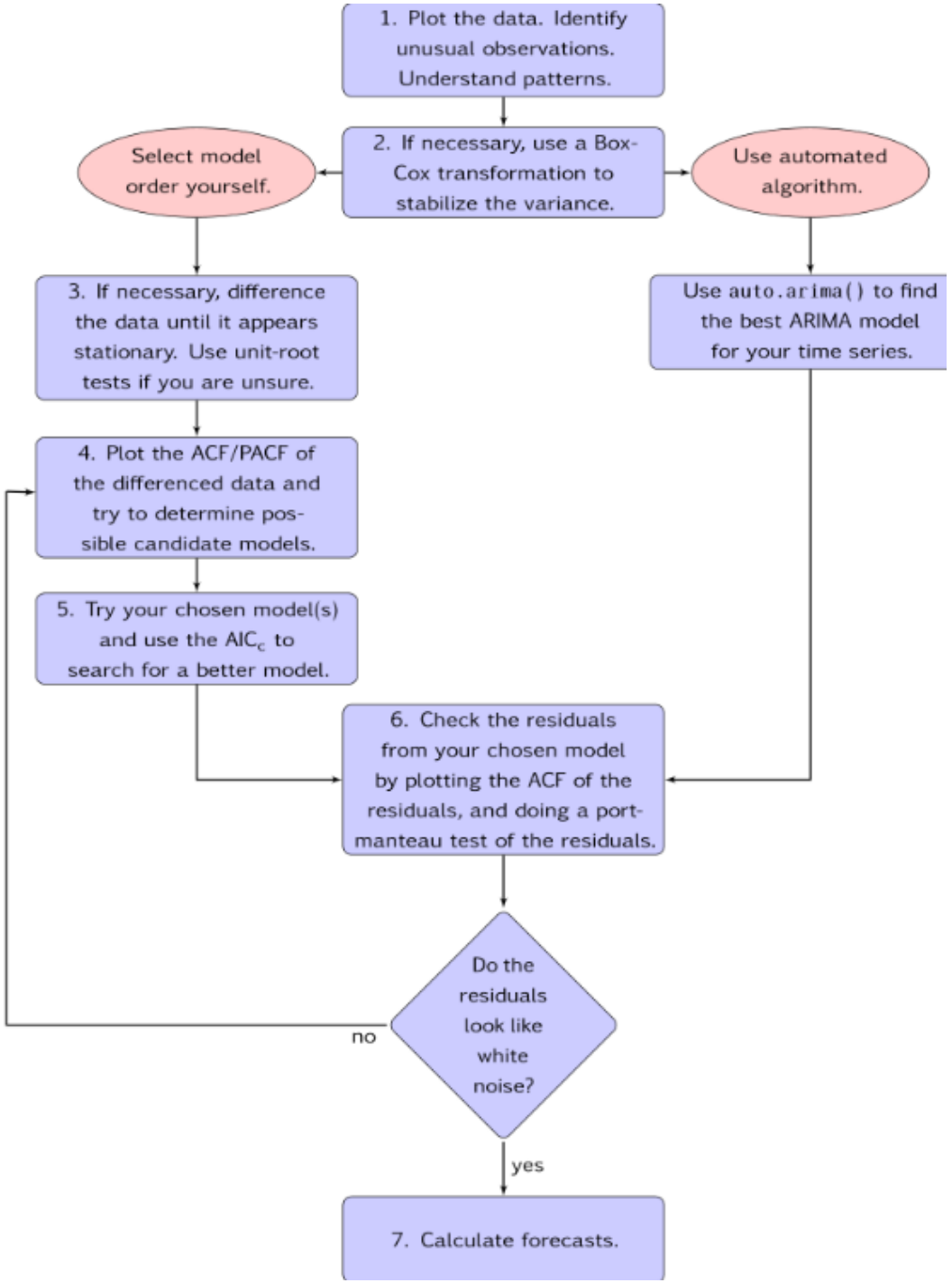
d. Repeat Step 2(c) until no lower AICc can be found.

Modeling Procedure

When fitting an ARIMA model to a set of (non-seasonal) time series data, the following procedure provides a useful general approach.

1. Plot the data and identify any unusual observations.
2. If necessary, transform the data (using a Box-Cox transformation) to stabilise the variance.
3. If the data are non-stationary, take first differences of the data until the data are stationary.
4. Examine the ACF/PACF: Is an ARIMA(p,d, 0) or ARIMA(0,d,q) model appropriate?
5. Try your chosen model(s), and use the AICc to search for a better model.
6. Check the residuals from your chosen model by plotting the ACF of the residuals, and doing a portmanteau test of the residuals. If they do not look like white noise, try a modified model.
7. Once the residuals look like white noise, calculate forecasts.

Modeling Process



Forecasting

Point forecast-Point forecasts can be calculated using the following **three** steps.

- Expand the ARIMA equation so that yt is on the left hand side and all other terms are on the right.
- Rewrite the equation by replacing t with T+h.
- On the right hand side of the equation, replace future observations with their forecasts, future errors with zero, and past errors with the corresponding residuals.

Beginning with h=1, these steps are then repeated for h=2,3,… until all forecasts have been calculated.

Point forecast Process- Step 1

- illustrate it using the ARIMA(3,1,1) model fitted

$$(1-\hat{\phi}_1 B-\hat{\phi}_2 B^2-\hat{\phi}_3 B^3)(1-B)y_t=(1+\hat{\theta}_1 B)\varepsilon_t,$$

where $\hat{\phi}_1=0.0044$, $\hat{\phi}_2=0.0916$, $\hat{\phi}_3=0.3698$ and $\hat{\theta}_1=-0.3921$. Then we expand the left hand side to obtain

$$\left[1-(1+\hat{\phi}_1)B+(\hat{\phi}_1-\hat{\phi}_2)B^2+(\hat{\phi}_2-\hat{\phi}_3)B^3+\hat{\phi}_3 B^4\right]y_t=(1+\hat{\theta}_1 B)\varepsilon_t,$$

and applying the backshift operator gives

$$y_t-(1+\hat{\phi}_1)y_{t-1}+(\hat{\phi}_1-\hat{\phi}_2)y_{t-2}+(\hat{\phi}_2-\hat{\phi}_3)y_{t-3}+\hat{\phi}_3 y_{t-4}=\varepsilon_t+\hat{\theta}_1\varepsilon_{t-1}.$$

Finally, we move all terms other than y_t to the right hand side:

$$y_t=(1+\hat{\phi}_1)y_{t-1}-(\hat{\phi}_1-\hat{\phi}_2)y_{t-2}-(\hat{\phi}_2-\hat{\phi}_3)y_{t-3}-\hat{\phi}_3 y_{t-4}+\varepsilon_t+\hat{\theta}_1\varepsilon_{t-1}.$$

Point forecast Process- Step 2

- For second step, replace t with T+1 in previous equation.

$$y_{T+1} = (1+\hat{\phi}_1)y_T - (\hat{\phi}_1 - \hat{\phi}_2)y_{T-1} - (\hat{\phi}_2 - \hat{\phi}_3)y_{T-2} - \hat{\phi}_3 y_{T-3} + \varepsilon_{T+1} + \hat{\theta}_1 \varepsilon_T.$$

Assuming we have observations up to time T, all values on the right hand side are known except for ε_{T+1}, which we replace with zero, and ε_T, which we replace with the last observed residual e_T:

$$\hat{y}_{T+1|T} = (1+\hat{\phi}_1)y_T - (\hat{\phi}_1 - \hat{\phi}_2)y_{T-1} - (\hat{\phi}_2 - \hat{\phi}_3)y_{T-2} - \hat{\phi}_3 y_{T-3} + \hat{\theta}_1 e_T.$$

Point forecast Process- Step 2

A forecast of y_{T+2} is obtained by replacing t with $T+2$ in (8.6) . All values on the right hand side will be known at time T except y_{T+1} which we replace with $\hat{y}_{T+1|T}$, and ε_{T+2} and ε_{T+1}, both of which we replace with zero:

$$\hat{y}_{T+2|T} = (1+\hat{\phi}_1)\hat{y}_{T+1|T} - (\hat{\phi}_1 - \hat{\phi}_2)y_T - (\hat{\phi}_2 - \hat{\phi}_3)y_{T-1} - \hat{\phi}_3 y_{T-2}.$$

The process continues in this manner for all future time periods. In this way, any number of point forecasts can be obtained.

Prediction intervals

- It is the range of values within which we expect the forecasted value to be.
- **Width of Prediction intervals** for non-time-series data is consistent throughout span of independent variable as the predictions are made within the range of values that the model was trained on.
- spread of Prediction intervals depends on forecasting horizon, further you try and predict into the future, **the wider the range** of prediction intervals

Ljung–Box test(1)

- The Ljung-Box statistic, also called the modified Box-Pierce statistic, is a function of the accumulated **sample autocorrelations, r_j**, up to any specified time **lag m**.
- As a function of m, it is determined as:

$$Q(m) = n(n+2)\sum_{j=1}^{m} \frac{r_j^2}{n-j},$$

where *n* = number of usable data points after any differencing operations.

Example:

$$Q(3) = n(n+2)\left(\frac{r_1^2}{n-1} + \frac{r_2^2}{n-2} + \frac{r_3^2}{n-3}\right)$$

Ljung–Box test(2)

- This statistic is used to examine residuals from a time series model
- used to see if all underlying population **autocorrelations** for errors may be **0** (up to a specified point).
- For all models residuals are assumed to be “white noise,” meaning that they are **identically, independently distributed** (from each other).
- Ideal ACF for residuals is that all autocorrelations are 0.
- This means that **Q(m)** should be 0 for any lag m .
- A significant **Q(m)** for residuals indicates a possible problem with the model.

Seasonal ARIMA models

- A seasonal ARIMA model is formed by including additional seasonal terms in ARIMA.
- It is written as follows:

$$\text{ARIMA} \quad \underbrace{(p, d, q)}_{\substack{\uparrow \\ \text{Non-seasonal part} \\ \text{of the model}}} \quad \underbrace{(P, D, Q)_m}_{\substack{\uparrow \\ \text{Seasonal part of} \\ \text{of the model}}}$$

- where m= number of observations per year.
- Use uppercase notation for the seasonal parts of model
- lowercase notation for the non-seasonal parts of model.

Seasonal ARIMA models

- Seasonal part of the model consists of terms that are similar to the non-seasonal components of the model, but involve backshifts of seasonal period.
- Example- an **ARIMA(1,1,1)(1,1,1)$_4$** model (without a constant) is for quarterly data (**m=4**), and can be written as

$$(1-\phi_1 B)\,(1-\Phi_1 B^4)(1-B)(1-B^4)y_t = (1+\theta_1 B)\,(1+\Theta_1 B^4)\varepsilon_t.$$

ACF/PACF

Seasonal part of an AR or MA model will be seen in **seasonal lags** of the PACF and ACF.

For example, an **ARIMA(0,0,0)(0,0,1)12** model will show:

- a spike at lag 12 in the ACF but no other significant spikes;
- exponential decay in the seasonal lags of the PACF (i.e., at lags 12, 24, 36, …).

Similarly, **an ARIMA(0,0,0)(1,0,0)12** model will show:

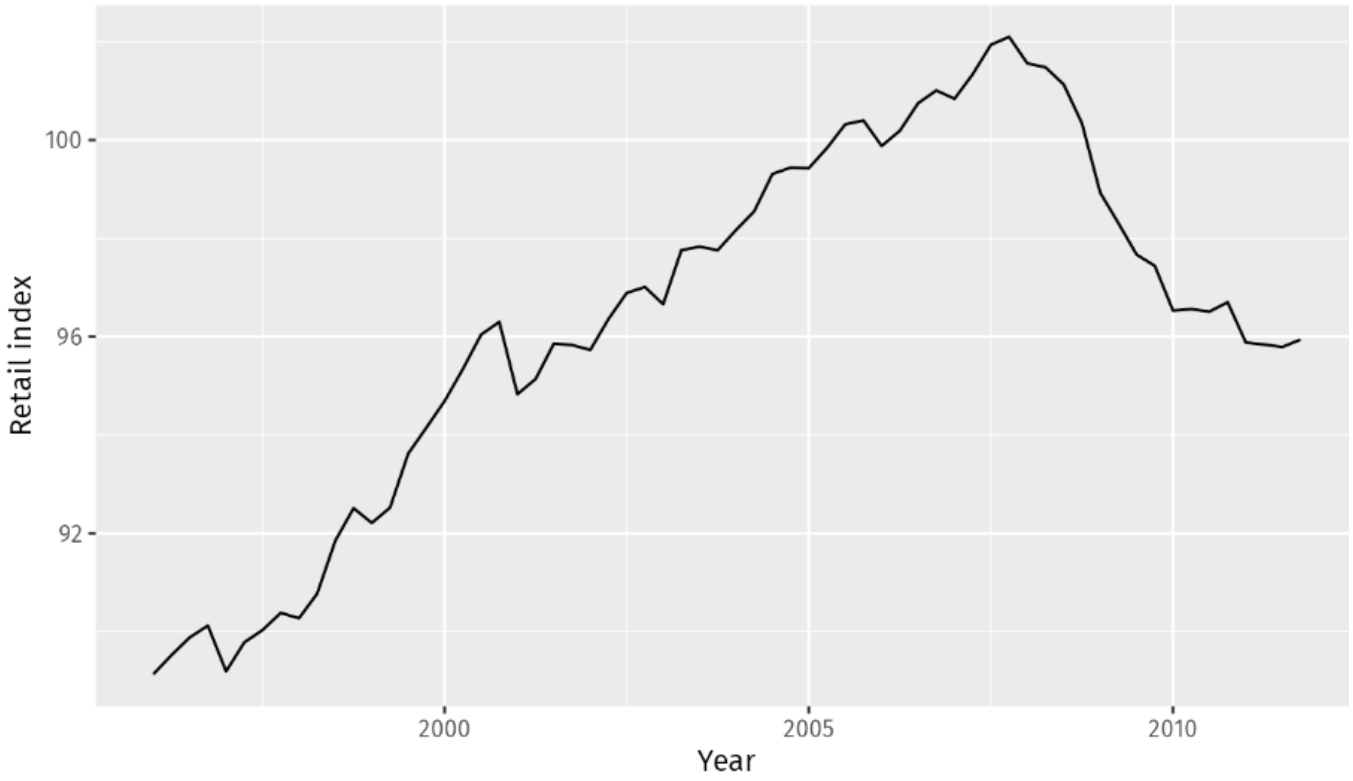
- exponential decay in the seasonal lags of the ACF;
- a single significant spike at lag 12 in the PACF.

In considering **seasonal orders** for a seasonal ARIMA model, restrict attention to seasonal lags.

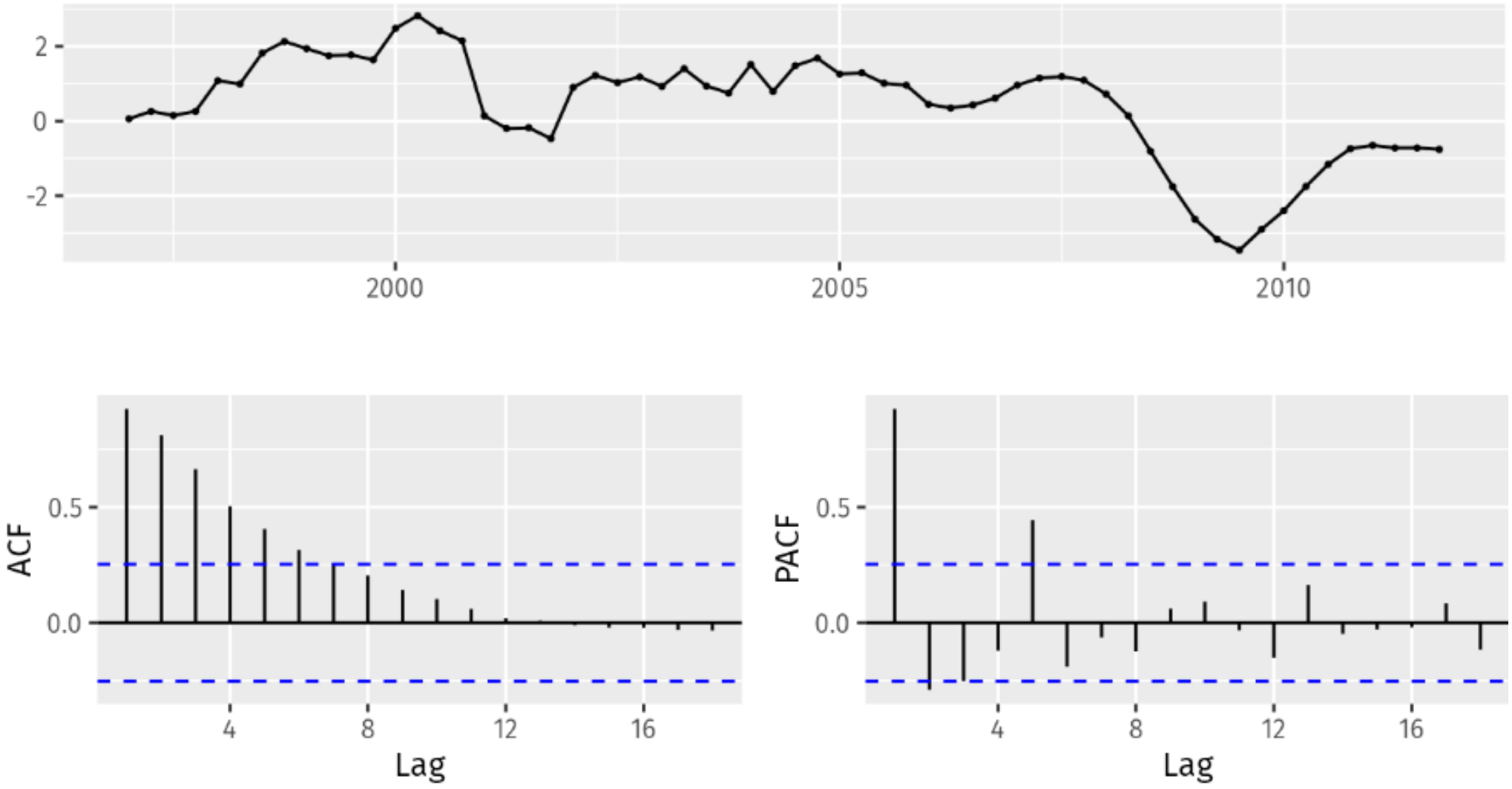
Modelling procedure is almost same as for non-seasonal data, except that we need to select seasonal AR and MA terms as well as the non-seasonal components of the model.

Example: European quarterly retail trade

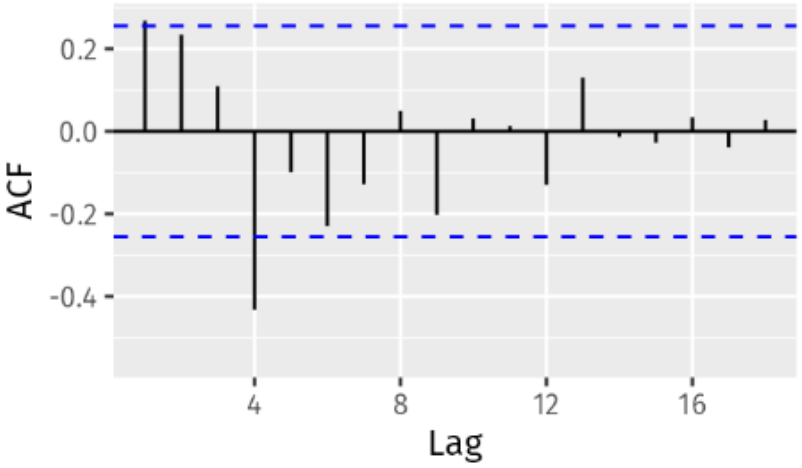
- Quarterly retail trade index in the Euro area (17 countries), 1996–2011
- Covers wholesale and retail trade, and the repair of motor vehicles and motorcycles. (Index: 2005 = 100).
- data are clearly non-stationary, with some seasonality, so
- take a seasonal difference.

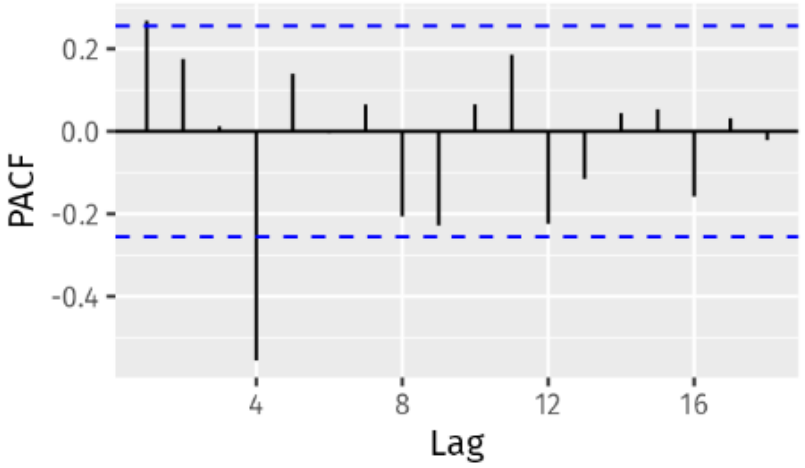


Example: European quarterly retail trade after seasonal difference



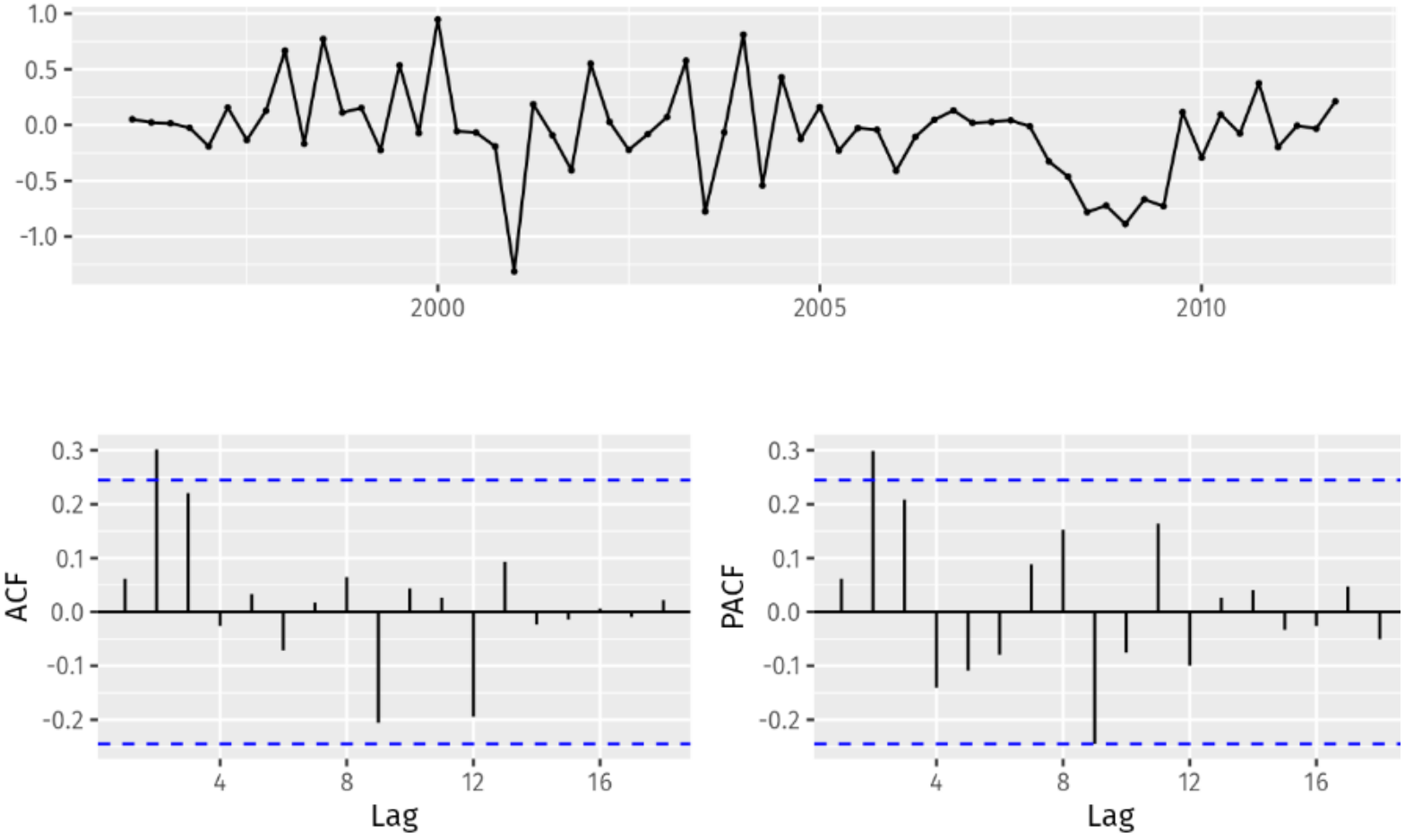
Example: European quarterly retail trade after one additional seasonal difference





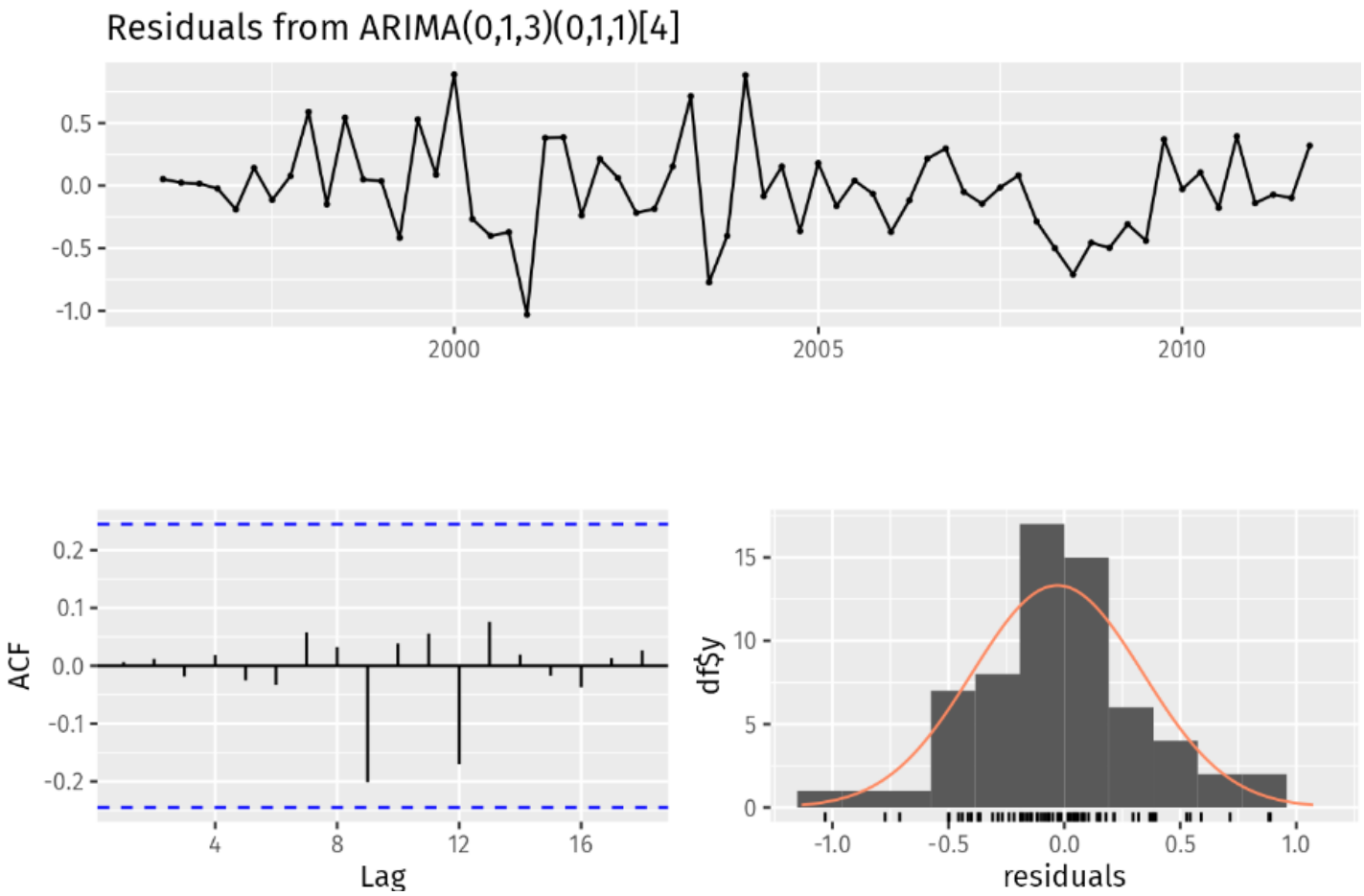
- spike at lag 1 in the ACF suggests a non-seasonal MA(1) component
- significant spike at lag 4 in ACF suggests a seasonal MA(1) component.
- So, begin with an **ARIMA(0,1,1)(0,1,1)$_4$** model, indicating a first and seasonal difference, and non-seasonal and seasonal MA(1) components.

Example: Residual for ARIMA(0,1,1)(0,1,1)$_4$



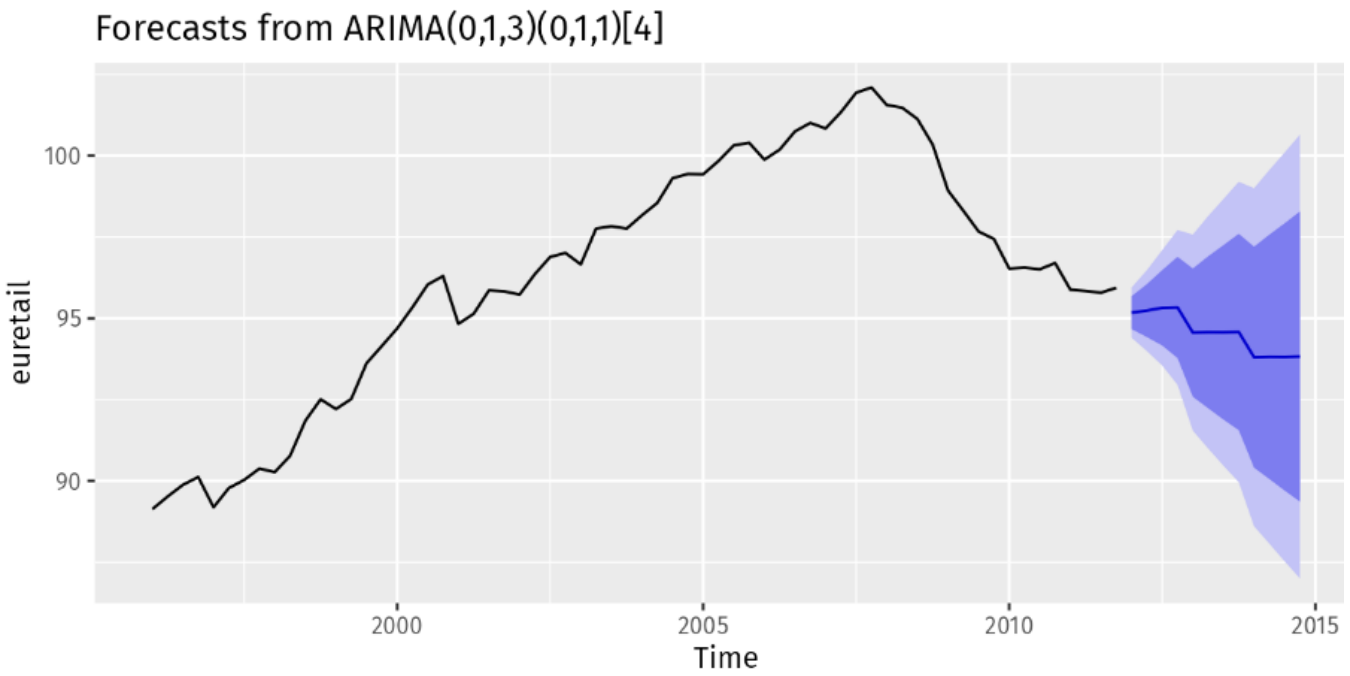
- Both ACF and PACF show significant spikes at lag 2, and spikes at lag 3.
- indicates that some additional non-seasonal terms need to be included in model.
- The AICc of **ARIMA(0,1,2)(0,1,1)$_4$** model is 74.36, while that for the **ARIMA(0,1,3)(0,1,1)$_4$** model is 68.53.
- Choose **ARIMA(0,1,3)(0,1,1)$_4$**

Example: Residual plot for ARIMA(0,1,3)(0,1,1)4



- All spikes are now within significance limits, so residuals appear to be white noise.
- Ljung-Box test shows that residuals have no remaining autocorrelations.
- This is a seasonal ARIMA model that **passes the required checks** and is ready for **forecasting**

Example: forecast for 3 years using ARIMA(0,1,3)(0,1,1)4



- forecasts follow recent trend in the data, because of the double differencing.
- large and **rapidly increasing prediction intervals** show that retail trade index could start increasing or decreasing at any time
- while point forecasts trend downwards, prediction intervals allow for data to trend upwards during the forecast period

ARIMAX (ARIMA with exogenous variables)

- ARIMAX is an extension of ARIMA model.
- In forecasting, this method involves independent variables also.
- ARIMAX model represents a composition of the output time series into the following parts:

  -autoregressive (AR) part
  -moving average (MA) part
  -integrated (I) component
  -part that belongs to the exogenous inputs

What is exogenous variable?

- A factor in the model whose value is identified by the factors of variables outside the model of study.
- **Example**- In farming, production forecast is based on factors like area of land, quality of seed etc. but in this rainfall is an exogenous variable whose value will be determined by other factors like humidity, air speed. Etc.

ARIMAX (ARIMA with exogenous variables)

- Referred to as the vector ARIMA or the dynamic regression model.
- It is similar to a multivariate regression model, but allows to take advantage of autocorrelation that may be present in residuals of the regression to improve the accuracy of a forecast.
- It includes also other independent (predictor) variables.

$$\Delta^D y_t = \sum_{i=1}^{p} \phi_i \Delta^D y_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} + \sum_{m=1}^{M} \beta_m Xm, t + \epsilon_t$$

$$\epsilon_t \sim N\left(0, \sigma^2\right)$$

Uses of ARIMAX

- Changes overtime is influenced by other factor.
- if we have sequentially data and predictors that influence it, we can use ARIMAX.
- Several use cases that has been done using ARIMAX:
  - -Quarterly changes in US consumption and personal income
  - -TV advertising and insurance quotations
  - -Pest Incidence of Cotton with Weather Factors
  - -Forecasting container throughput

Seasonal ARIMA with exogenous variables

- It is a powerful time series forecasting technique that extends traditional ARIMA model to account for **seasonality and external factors.**
- Accommodate both autoregressive (AR) and moving average (MA) components, integrate differencing to make data stationary, and incorporate external variables or regressors.
- **SARIMAX** is valuable when dealing with time-dependent data that exhibits recurring patterns over specific time

SARIMAX Components

- **Seasonal Component (S):** Captures the periodic patterns in the data, such as weekly, monthly, or yearly cycles.
- **Autoregressive Component (AR):** Represents the relationship between the current value and previous values in the time series.
- **Integrated Component (I):** Involves differencing to make the time series stationary by removing trends and seasonality.
- **Moving Average Component (MA):** Accounts for the dependency of the current value on past error terms, used to calculate trend.
- **Exogenous Regressors (X):** Allows the inclusion of external variables that may affect the time series.

Regression with ARIMA errors in R

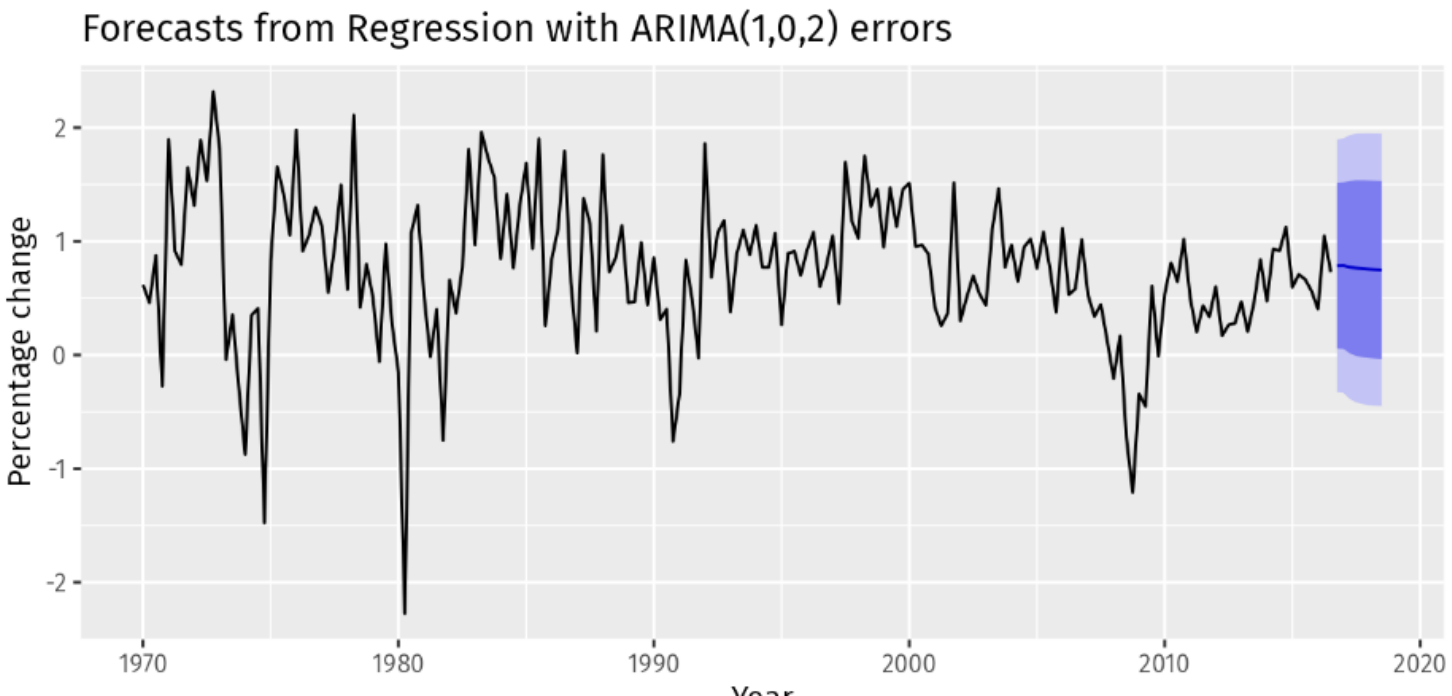
- R function **Arima()** will fit a regression model with ARIMA errors if the argument **xreg** is used.
- Order argument specifies order of ARIMA
- If differencing is used, then it is applied to all variables.

```
fit <- Arima(y, xreg=x, order=c(1,1,0))
```

Forecasting with Extra predictors

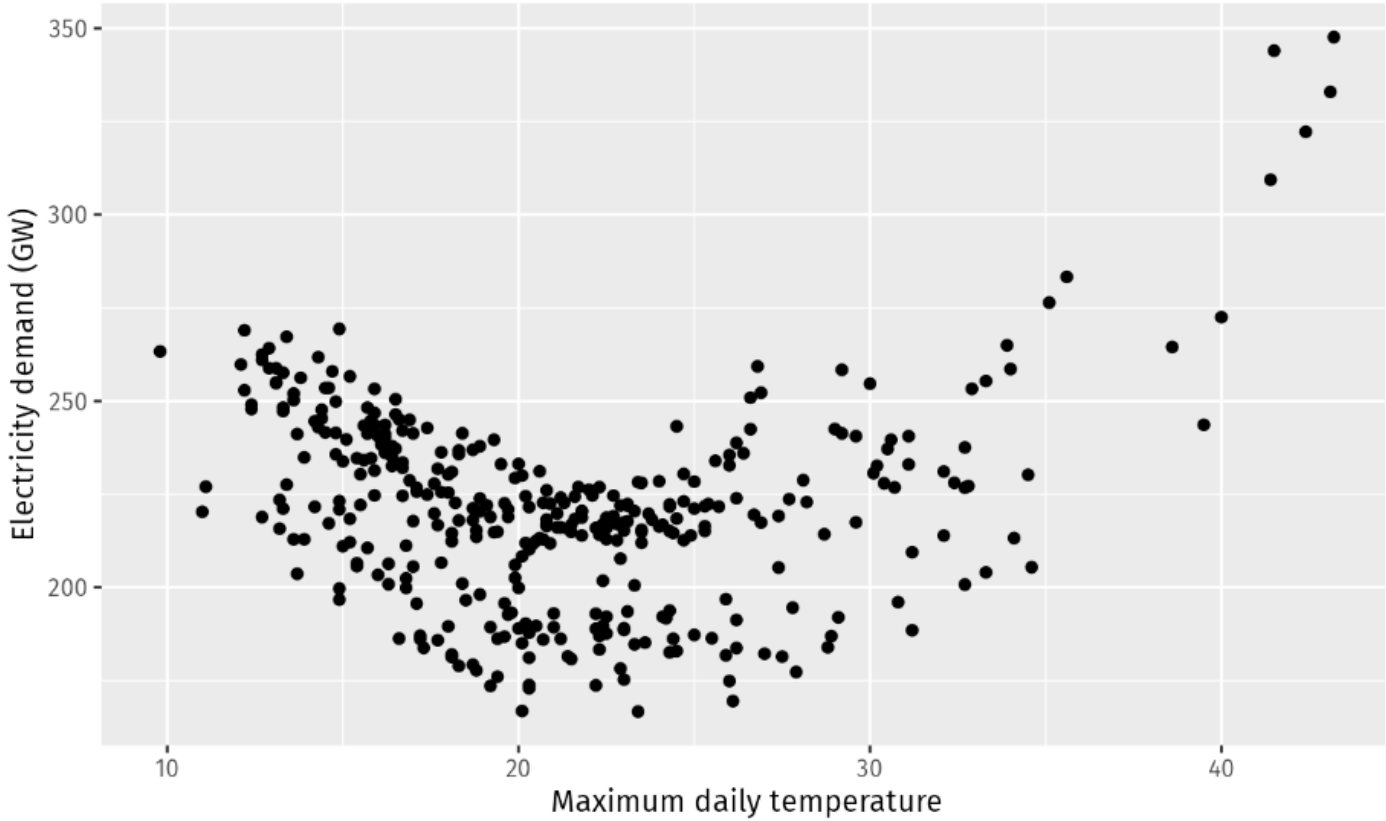
- calculate forecasts for **next eight quarters** assuming that future percentage changes in **personal disposable income** will be equal to mean percentage change from last forty years.

```
fcast <- forecast(fit, xreg=rep(mean(uschange[,2]),8))
autoplot(fcast) + xlab("Year") +
  ylab("Percentage change")
```



Example: Forecasting electricity demand

- Daily electricity demand can be modelled as a function of temperature.
- observed on an electricity bill, more electricity is used on cold days due to heating and hot days due to air conditioning.
- higher demand on cold and hot days is reflected in the u-shape of figure



Exercise

1. Consider monthly sales and advertising data for an automotive parts company (data set `advert`).

   a. Plot the data using `autoplot`. Why is it useful to set `facets=TRUE`?

   b. Fit a standard regression model $y_t = a + bx_t + \eta_t$ where y_t denotes sales and x_t denotes advertising using the `tslm()` function.

   c. Show that the residuals have significant autocorrelation.

   d. What difference does it make you use the `Arima` function instead:

   ```
   Arima(advert[,"sales"], xreg=advert[,"advert"],
     order=c(0,0,0))
   ```

   e. Refit the model using `auto.arima()`. How much difference does the error model make to the estimated parameters? What ARIMA model for the errors is selected?

   f. Check the residuals of the fitted model.

   g. Assuming the advertising budget for the next six months is exactly 10 units per month, produce and plot sales forecasts with prediction intervals for the next six months.

Exercise

2. This exercise uses data set `huron` giving the level of Lake Huron from 1875–1972.

   a. Fit a piecewise linear trend model to the Lake Huron data with a knot at 1920 and an ARMA error structure.

   b. Forecast the level for the next 30 years.